

Week 1 Report: Financial Data Acquisition and Exploratory Data Analysis (EDA)

Date: September 2026

1. Introduction & Executive Summary

Objective:

The primary goal of this report is to analyze historical financial data to uncover underlying market trends, price volatility, and trading behaviors. By conducting an Exploratory Data Analysis (EDA), we aim to establish a clean and robust dataset that will serve as the foundation for future financial modeling and forecasting.

Dataset Overview:

The data analyzed was sourced from the provided yahoo finance.xlsx file, which contains daily stock price metrics for Apple Inc. (AAPL). The scope of this dataset covers 254 trading days, tracking daily market activity from September 10, 2019, through September 10, 2020.

2. Data Acquisition & Description

Data Source:

The dataset was acquired from the yahoo finance.xlsx file (specifically the 'AAPL_daily_update' sheet), representing structured historical market data.

Data Dictionary:

Column Name	Data Type	Description
Date	Datetime	The specific trading date
Open	Numeric	Opening price of the asset for the day
High	Numeric	Highest price reached during the trading day
Low	Numeric	Lowest price reached during the trading day
Close	Numeric	Closing price of the asset at the end of the day
Adj Close	Numeric	Closing price adjusted for splits and dividends

Volume	Numeric	Total number of shares traded during the day
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3. Data Cleaning & Preprocessing

Missing Values:

A thorough check revealed zero missing or NaN values across all 254 records. Consequently, no rows required dropping or imputation (such as forward filling), ensuring complete data integrity.

Data Formatting:

The dataset was structurally sound. The Date column was successfully verified and formatted as a proper datetime data type. The price columns (Open, High, Low, Close, Adj Close) were confirmed as floating-point numbers, and Volume was confirmed as a 64-bit integer.

Outliers & Duplicates:

An analysis for duplicate entries returned zero duplicate rows. While no erroneous outliers were removed, extreme legitimate market spikes were identified, most notably a massive trading volume surge on March 12, 2020, where over 418 million shares were traded—likely corresponding to broad market volatility during the early pandemic period.

4. Exploratory Data Analysis (EDA)

Summary Statistics:

Metric	Open	High	Low	Close	Volume
Mean	77.81	78.88	76.85	77.96	144,632,300
Std Dev	18.39	18.63	17.87	18.24	67,353,290
Minimum	53.46	54.19	52.92	54.17	46,617,600
25th Percentile	64.83	65.80	64.53	65.45	100,625,900
50th Percentile	73.96	75.18	72.61	74.37	127,577,000
75th Percentile	85.46	86.81	83.45	85.48	167,681,200

Maximum	137.58	137.97	130.53	134.18	418,474,000
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Key Visualizations:

- **Line Chart:** Time-series trends would illustrate a strong overall upward trajectory in the asset's price, starting from a low of ~54.17 in September 2019 and peaking at ~134.18 by September 2020, disrupted by a noticeable dip in March 2020.
- **Bar Chart / Histogram:** Volume distributions show that while the median daily volume sits around 127 million shares, there is a distinct positive skew indicating periodic days of heavy panic or speculative trading.

Analysis & Insights:

The dataset reveals a highly bullish overall trend for the asset during the analyzed year, with the closing price increasing by nearly 150% from its minimum to maximum values. This growth, however, was not strictly linear. The standard deviation of ~18.24 on the closing price highlights significant volatility, matching broader global economic uncertainties during the 2019–2020 timeframe.

The trading volume provides a crucial secondary narrative. The massive volume spike observed on March 12, 2020 (over 418 million shares traded), aligns with a period of high market distress. The weak correlation between volume and closing price suggests that these massive sell-offs and buy-ins were driven by external market shocks rather than standard technical price progression.

Overall, the high correlation among intra-day prices (Open, High, Low, Close) indicates high daily price stability relative to the broader macro trends. The lack of missing data or anomalies means this dataset is highly reliable for continuing into predictive modeling.

5. Key Findings & Recommendations

Summary of Observations:

1. **Strong Upward Trend:** The asset demonstrated massive growth over the one-year period, effectively doubling in value despite macroeconomic volatility.
2. **Identifiable Volatility Spikes:** Extreme volume days (such as the 418M+ shares traded in March 2020) act as clear markers of market shock, which will need to be accounted for in future predictive models.
3. **Intra-day Price Consistency:** The near-perfect correlation (0.99+) between Open, High, Low, and Close prices means that any single price metric can serve as a highly reliable baseline indicator for daily performance without losing much context.

Next Steps:

Moving into Week 2, this cleaned dataset will form the foundation for predictive financial forecasting. The Close or Adj Close price will be selected as the primary target variable. Future work will involve engineering new features (such as 7-day and 30-day moving averages) and

testing initial regression models to predict future price movements based on historical patterns.